

Thursday Warm-up: Potentials

Prof. Jordan C. Hanson

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1 Memory Bank

- **Fourier series** for a function $f(x)$:

$$f(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos nx + B_n \sin nx) \quad (1)$$

$$A_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos(nx) dx \quad (n = 0, 1, 2, \dots) \quad (2)$$

$$B_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(nx) dx \quad (n = 1, 2, \dots) \quad (3)$$

- **Trigonometric identities**:

$$\cos \theta \cos \phi = \frac{1}{2} (\cos(\theta - \phi) + \cos(\theta + \phi)) \quad (4)$$

$$\sin \theta \sin \phi = \frac{1}{2} (\cos(\theta - \phi) - \cos(\theta + \phi)) \quad (5)$$

$$\sin \theta \cos \phi = \frac{1}{2} (\sin(\theta + \phi) + \sin(\theta - \phi)) \quad (6)$$

$$\cos \theta \sin \phi = \frac{1}{2} (\sin(\theta + \phi) - \sin(\theta - \phi)) \quad (7)$$

- **Laplace's equation** in spherical coordinates, with azimuthal symmetry:

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) = 0 \quad (8)$$

2 Separation of Variables

1. **Fourier Series.** Let $\delta_{nn'} = 1$, if $n = n'$, and 0 if $n \neq n'$.
(a) Show that the cosine functions are *orthogonal*, using an identity in the Memory Bank. That is, prove that

$$\int_0^a \cos(n\pi x/a) \cos(n'\pi x/a) dx = \frac{a}{2} \delta_{nn'} \quad (9)$$

(b) Create the Fourier series for a *triangle* wave. That is, let $f(x) = x$ if $0 \leq x \leq \pi$, and $f(x) = 2\pi - x$ if $\pi \leq x < 2\pi$. Draw a graph of the *spectrum*.

2. A hollow spherical shell of radius R has a potential $V_0(\theta) = V_0$ for $\theta \leq \pi/2$, and 0 for $\theta > \pi/2$. Find the voltage outside the shell.

- (a) Separate the solution for the voltage like $V(r, \theta) = R(r)\Theta(\theta)$. Isolate the $R(r)$ and r factors from the Θ and θ factors. (b) Set the ordinary differential equation for $R(r)$ equal to the constant $l(l+1)$, where l is an integer. Show that the solutions are linear combinations of r^l and $r^{-(l+1)}$.

- The $\Theta(\theta)$ solutions are known as *Legendre polynomials*, $P_l(x)$, and $x = \cos \theta$, with the **Rodrigues** generating function

$$P_l(x) = \frac{1}{2^l l!} \left(\frac{d}{dx} \right)^l (x^2 - 1)^l \quad (10)$$

Derive $P_0(x)$, $P_1(x)$, and $P_2(x)$.

- Write the solution for $V(r, \theta)$ as an infinite sum, and apply the boundary condition for $r \rightarrow \infty$.

- Apply the boundary condition at $r = R$, and use Fourier's trick to solve for the coefficients. You will need the following orthogonality condition:

$$\int_{-1}^1 P_l(x) P_l'(x) dx = \int_0^\pi P_l(\cos \theta) P_l'(\cos \theta) \sin \theta d\theta = \frac{2}{2l+1} \delta_{ll'} \quad (11)$$